

Experiment Name: 01

Name of the Experiment: Familiarization with Electrical Circuits Lab equipment, measurement the resistance according to color codes and verification of Ohm's law.

Objective:

The objective of the experiment is to learn about the commonly used equipment in the lab and how to properly use it. To verify the following two equivalent forms of Ohm's Law:

- Express I as a function of V and R .
- Express V as a function of I and R .

Breadboard

Breadboard is a board used for electrical circuit prototyping. Before the circuit is permanently placed in a PCB (Printed Circuit Board) prototyping boards are used to verify the electrical circuit. Some other prototyping boards are: Perf board, Strip board, Vero board etc.

Top two row (A and B) of the board are internally connected sideways and the holes in group B are connected vertically as show in the figure 1 and this sequence continue in the rest of the board.

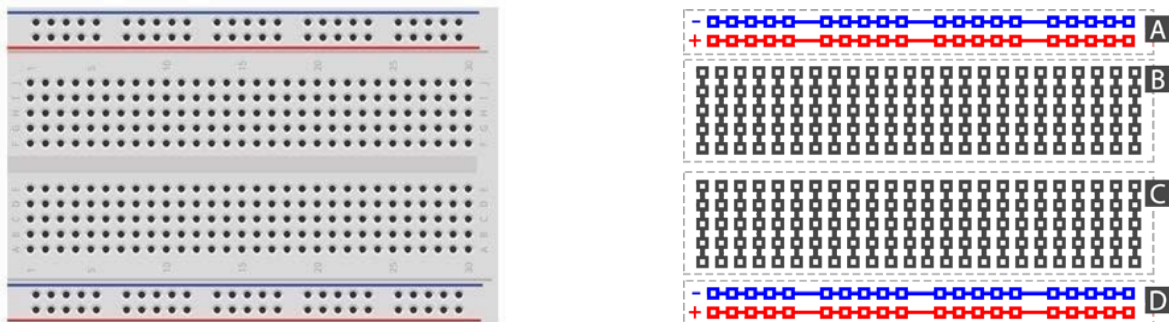


Figure 1.1: External and Internal construction of the board

Breadboard From circuit to Breadboard:

Closely observe the circuit diagram (electronic schematic) and the equivalent connection on breadboard.

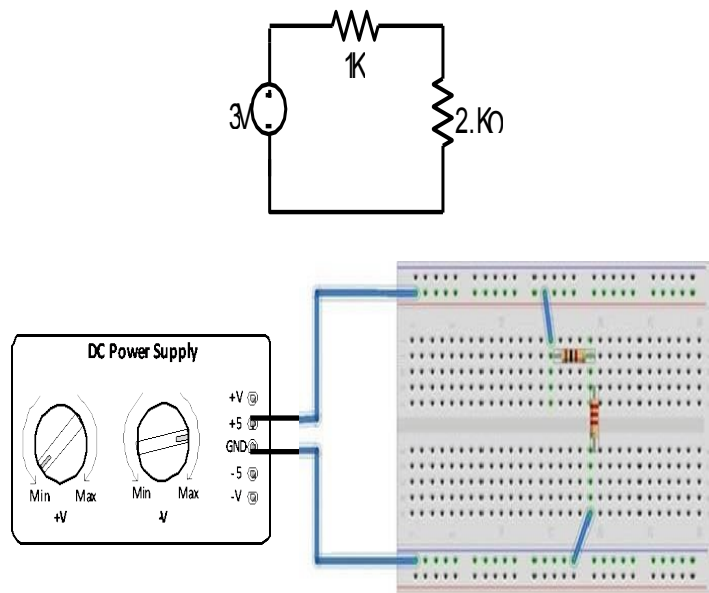
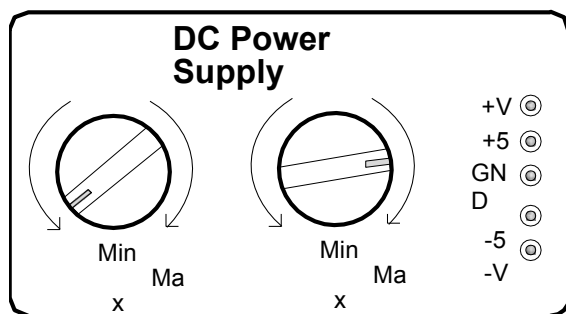


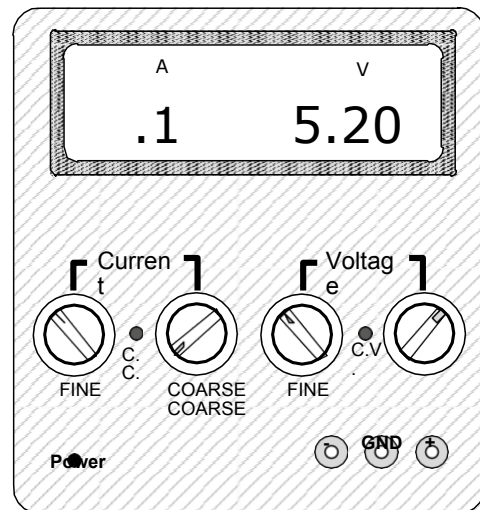
Figure 1.2: Circuit diagram and the equivalent breadboard connection

DC Power Supply:

In the Lab we have two DC source in the workbench. One can be found in the trainer board Fig 1.3(a) and another is a individual DC power supply module Fig 1.3(b).



(a) Trainer board DC source



(b) DC power supply

Figure 1.3: DC power sources

Trainer board DC source:

Trainer board DC power supply can deliver two variable and two fixed DC voltage at the same time. The variable voltage can be adjusted using the two dials.

Voltage output at pin (Reference to GND)	Value	Type
+V	1.2 V to 20 V	Variable
+5	5 V	Fixed
-5	-5 V	Fixed
-V	-1.2 V to 20V	Variable

DC power supply module:

The DC power supply module can deliver voltage ranging from 0V to 30V.

1. To set a particular voltage turn on the power supply.
2. To change the voltage in big step use the coarse dial and to precisely vary the voltage use the fine dial in the voltage group.
3. Observe the output voltage change.

The course and fine dial in current group is used to set the maximum current limit at the output.

Measuring Voltage:

1. To measure the voltage across the $1K\Omega$ resistor circuit in Fig 1.4 (a), construct the circuit as shown in Fig 1.4 (b).
2. Rotate the multimeter dial in the V position.
3. Connect the red and black multimeter lead as shown in the Fig 1.4(a) (parallel to the resistor)
4. Multimeter should display the Voltage.

Warning: while measuring voltage multimeter dial SHOULD NOT be in mA (current) position. It might destroy/damage the meter.

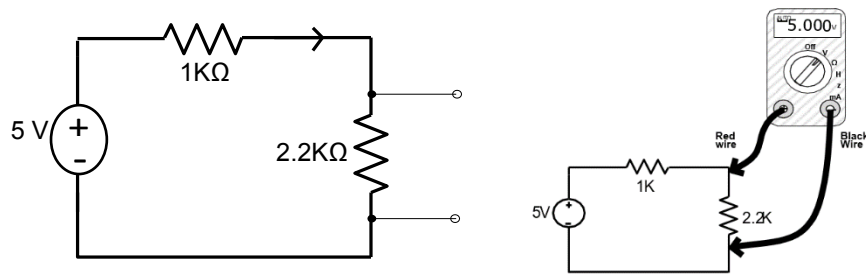


Figure 1.4: Circuit diagram and actual connection for measuring voltage

Measuring Current:

- 1.To measure the current in the series circuit in Fig 1.5 construct the circuit and then create a break the circuit as shown in Fig 1.5
- 2.Rotate the multimeter dial in the mA position.
- 3.Connect the red and black multimeter lead as shown in the Fig 1.5 (in series with the circuit)
- 4.Multimeater should display the current.

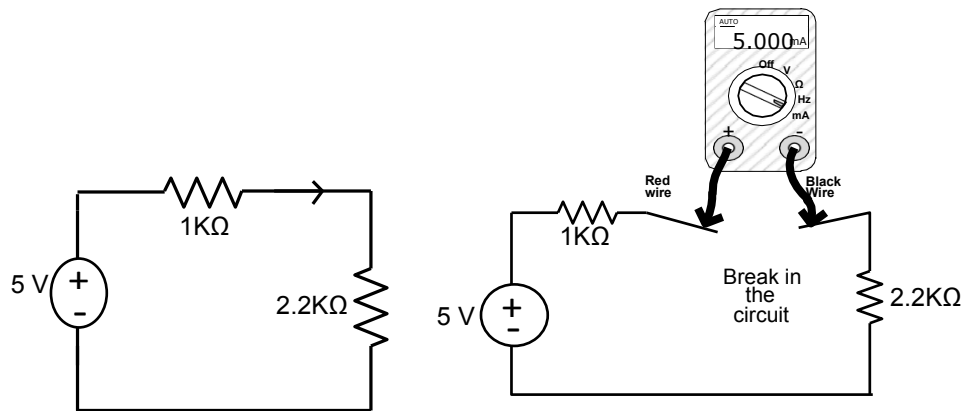


Figure 1.5: Circuit diagram and actual connection for measuring current

Resistors:

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element.

The current through a resistor is in direct proportion to the voltage across the resistor's terminals. This relationship is represented by Ohm's law: $I = V/R$, where I is the current through the conductor in units of

amperes, V is the potential difference measured across the conductor in units of volts, and R is the resistance of the conductor in units of ohms.

The ratio of the voltage applied across a resistor's terminals to the intensity of current in the circuit is called its resistance, and this can be assumed to be a constant (independent of the voltage) for ordinary resistors working within their ratings.

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel-chrome). Resistors are also implemented within integrated circuits, particularly analog devices, and can also be integrated into hybrid and printed circuits.

The electrical functionality of a resistor is specified by its resistance: common commercial resistors are manufactured over a range of more than nine orders of magnitude. When specifying that resistance in an electronic design, the required precision of the resistance may require attention to the manufacturing tolerance of the chosen resistor, according to its specific application. The temperature coefficient of the resistance may also be of concern in some precision applications. Practical resistors are also specified as having a maximum power rating which must exceed the anticipated power dissipation of that resistor in a particular circuit: this is mainly of concern in power electronics applications. Resistors with higher power ratings are physically larger and may require heat sinks. In a high-voltage circuit, attention must sometimes be paid to the rated maximum working voltage of the resistor.

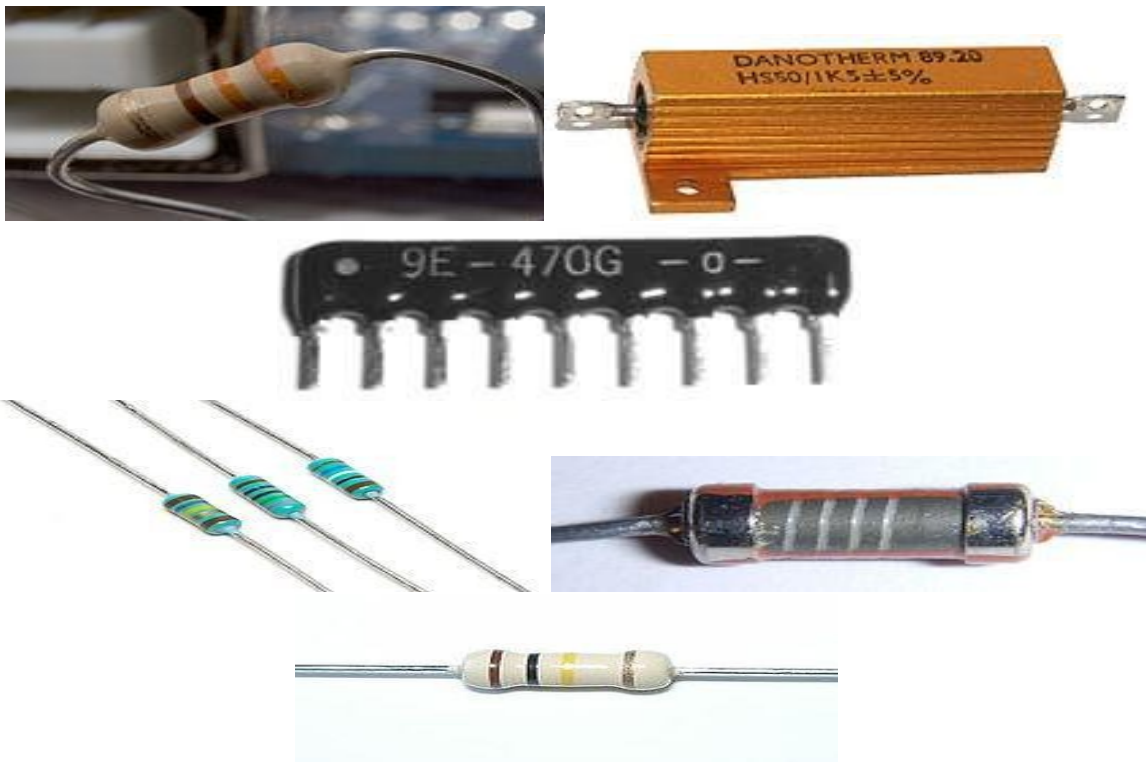


Figure 1.6: Different types of resistors

Practical resistors have a series inductance and a small parallel capacitance; these specifications can be important in high-frequency applications. In a low-noise amplifier or pre-amp, the noise characteristics of a resistor may be an issue. The unwanted inductance, excess noise, and temperature coefficient are mainly dependent on the technology used in manufacturing the resistor. They are not normally specified individually for a particular family of resistors manufactured using a particular technology. A family of discrete resistors is also characterized according to its form factor, that is, the size of the device and the position of its leads (or terminals) which is relevant in the practical manufacturing of circuits using them.

Resistor Color Coding:

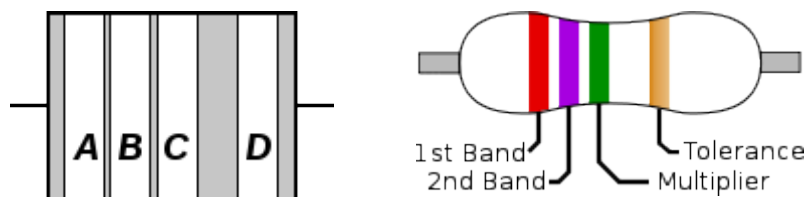
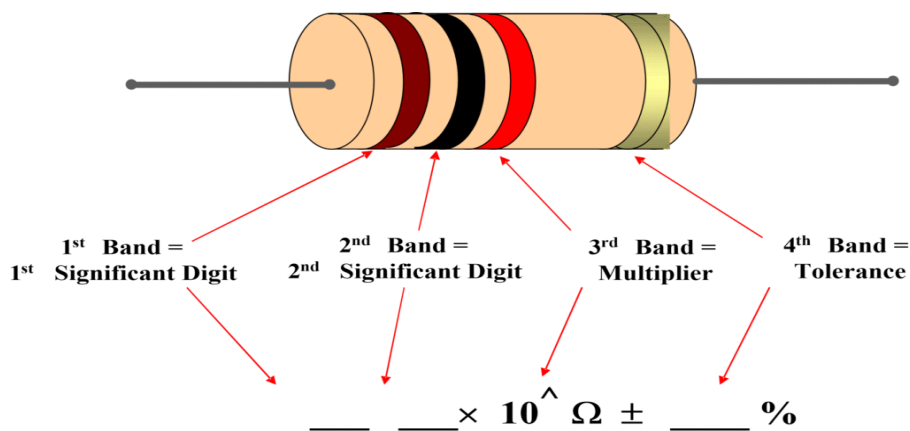


Figure 1.7: Resistor color coding

A 2260-ohm, 1% precision resistor with 5 color bands (E96 series), from top 2-2-6- 1-1; the last two brown bands indicate the multiplier (x10), and the 1% tolerance. The larger gap before the tolerance band is somewhat difficult to distinguish. To distinguish left from right there is a gap between the C and D bands.

- band **A** is first significant figure of component value (left side)
- band **B** is the second significant figure (Some precision resistors have a third significant figure, and thus five bands.
- band **C** is the decimal multiplier.
- band **D** if present, indicates tolerance of value in percent (no band means 20%).



For example, a resistor with bands of yellow, violet, red, and gold will have first digit 4.

(yellow in table below), second digit 7 (violet), followed by 2 (red) zeros: 4,700 ohms. Gold signifies that the tolerance is $\pm 5\%$, so the real resistance could lie anywhere between 4,465 and 4,935 ohms.

Resistors manufactured for military use may also include a fifth band which indicates component failure rate (reliability); refer to MIL-HDBK-199 for further details.

Tight tolerance resistors may have three bands for significant figures rather than two, or an additional band indicating temperature coefficient, in units of ppm/K.

All coded components will have at least two value bands and a multiplier; other bands are optional.

The standard color code per EN 60062: 2005 is as follows:

Color		Significant figures		Multiplier		Tolerance	Temp. Coefficient (ppm/K)
Black		0		×10 ⁰		—	250
Brown		1		×10 ¹		±1%	F
Red		2		×10 ²		±2%	G
Orange		3		×10 ³		—	15
Yellow		4		×10 ⁴		(±5%)	—
Green		5		×10 ⁵		±0.5%	D
Brown		1		×10 ¹		±1%	F
Blue	6	×10 ⁶	±0.25 %	C	10	Z	
Violet	7	×10 ⁷	±0.1 %	B	5	M	
Gray	8	×10 ⁸	±0.05 % (±10 %)	A	1	K	
White	9	×10 ⁹	—		—		
Gold	—	×10 ⁻¹	±5%	J	—		
Silver	—	×10 ⁻²	±10 %	K	—		
None	—	—	±20 %	M	—		

1. Any temperature co-efficient not assigned its own letter shall be marked "Z", and the coefficient found in other documentation.
2. For more information, see EN 60062.
3. Yellow and Gray are used in high-voltage resistors to avoid metal particles in the lacquer.

Resistors use preferred numbers for their specific values, which are determined by their tolerance. These values repeat for every decade of magnitude: 6.8, 68, 680, and so forth. In the E24 series the values are related by the 24th root of 10, while E12 series are related by the 12th root of 10, and E6 series by the 6th root of 10. The tolerance of device values is arranged so that every value corresponds to a preferred number, within the required tolerance.

Zero-ohm resistors are made as lengths of wire wrapped in a resistor-shaped body which can be substituted for another resistor value in automatic insertion equipment. They are marked with a single black band.

The 'body-end-dot' or 'body-tip-spot' system was used for radial-lead (and other cylindrical) composition resistors sometimes still found in very old equipment; the first band was given by the body color, the second band by the color of the end of the resistor, and the multiplier by a dot or band around the middle of the resistor. The other end of the resistor was colored gold or silver to give the tolerance, otherwise it was 20%.

Verification of Ohm's Law

THEORY:

Ohm's law describes mathematically how voltage 'V', current 'I' and resistance 'R' in a circuit are related. According to this law:

“The current in a circuit is directly proportional to the applied voltage and inversely proportional to the circuit resistance”.

Formula for voltage:

For a constant value of R, V is directly proportional to I

i.e. $V = IR$

Formula for current:

For a constant value of V, I is inversely proportional to R

i.e. $I = V/R$

EQUIPMENT:

- Variable DC power supply -1 piece
- Digital multimeter (DMM)/ Analog multimeter-1 piece.
- Resistances: 1 K Ω , 2.2 K Ω , 3.3 K Ω , 4.7 K Ω , 5.6 K Ω , 10 K Ω -1 piece each.
- Trainer Board
- Connecting Wires.

CIRCUIT DIAGRAM:

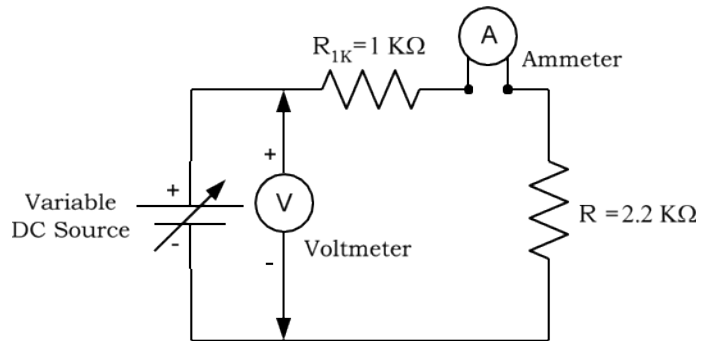


Figure 1.8: Verification of Ohm's Law

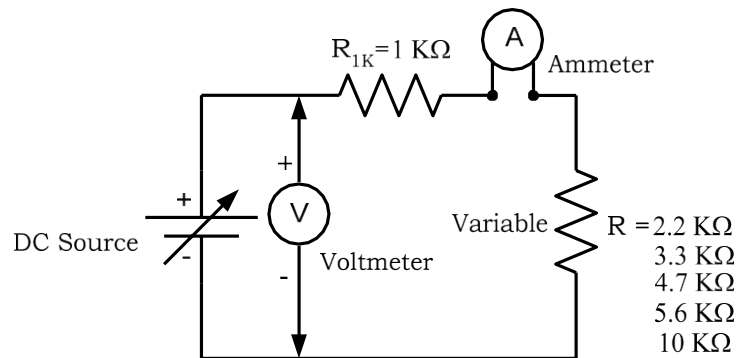


Figure 1.9: Verification of Ohm's Law

PROCEDURES:

Current versus voltage:

- Construct the circuit of Figure 1.8. Do not switch on the power supply.
- Turn on the power supply and adjust it to 5V by using Voltmeter. Measure the current I by ammeter and record it in the Table 1.9.
- Increase the values of voltage as shown in the Table 1.9. Measure the current I in turn and record the values in Table 1.9.
- Calculate the values of current I by using $I = V/RT$. Use measured values of resistances.

Current versus resistance:

- Construct the circuit of Figure 1.9. Do not switch on the power supply.
- Turn on the power supply and adjust it to 20V by using Voltmeter. Measure the current I by ammeter for $R=2.2\text{ K}\Omega$ (Use measured values) and record it in the Table 1.3.
- Turn off the power supply and remove the resistance $2.2\text{ K}\Omega$. Replace it by resistor $3.3\text{ K}\Omega$.
- Now turn on the power supply. Measure and record the current I in turn, at each of the resistance settings shown in the Figure 1.2.
- Calculate the values of resistance R_T by using $R_T=V/I$. Use measured values of voltage and current.

DATA SHEET

Table 1.1: Measuring Resistances by using Ohmmeter

Nominal values of $R\text{ (K}\Omega\text{)}$	Measured values of $R\text{ (K}\Omega\text{)}$ by using Ohmmeter
1	
2.2	
3.3	
4.7	
5.6	
10	

Table 1.2: Current versus voltage

Supply Voltage(V)	Measured I by using Ammeter (A)	$R_T = R_{1K} + R_{2.2K}$ [Use measured values of R]	Calculate I (amp) $I = V/R_T$
5			
10			
15			
20			
25			

Table 1.3: Current versus resistance

Supply Voltage (V)	Measured I by using Ammeter (A)	$R_T (K\Omega)$ Use measured values of R	Calculate $R_T = V/I (K\Omega)$
20		$R_T = R_{1K} + R_{2.2K}$	
		$R_T =$	
20		$R_T = R_{1K} + R_{3.3K}$	
		$R_T =$	
20		$R_T = R_{1K} + R_{4.7K}$	
		$R_T =$	
20		$R_T = R_{1K} + R_{5.6K}$	
		$R_T =$	
20		$R_T = R_{1K} + R_{10K}$	
		$R_T =$	

ASSIGNMENTS:

1. What can you say about the relationship between voltage and current, provided that the resistance is fixed?
2. Plot a graph of I versus V keeping the value of resistance constant. Use measured values of I and V. Comment on the graph briefly.
3. Plot a graph of I versus R_T keeping the value of supply voltage constant. Use measured values of I and R_T . Comment on the graph briefly.

Experiment No: 02

Experiment Name: Verification of KVL and KCL in DC circuits

OBJECTIVE:

The objective of this experiment is to investigate the characteristics of a series DC circuit and to verify Kirchoff's Voltage Law (KVL) and parallel DC circuit and to verify Kirchoff's Current Law (KCL).

THEORY:

In a series circuit (Figure 3.1) the current is same through all of the circuit elements. The equivalent Resistance, $R_T = R_1 + R_2 + R_3$.

By Ohm's Law, the Current is

$$I = \frac{V_{\text{Supply}}}{R_T}$$

KVL states that the voltage rises must be equal to the voltage drops around a close circuit. Applying Kirchoff's Voltage Law around closed loop of Figure 3.1, we find,

$$V_{\text{supply}} = V_1 + V_2 + V_3$$

Where, $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$.

Current I is same throughout the circuit for figure 2.1.

The voltage divider rule states that the voltage across an element or across a series combination of elements in a series circuit is equal to the resistance of the element divided by total resistance of the series circuit and multiplied by the total impressed voltage. For the elements of Figure 2.1.

$$V_1 = \frac{R_1 V_{\text{supply}}}{R_T} \quad V_2 = \frac{R_2 V_{\text{supply}}}{R_T} \quad V_3 = \frac{R_3 V_{\text{supply}}}{R_T}$$

EQUIPMENT:

- Variable DC power supply -1 piece
- Digital Multimeter (DMM)/ Analog multimeter-1 piece.
- Resistances: 100Ω, 220Ω, 470Ω -1 piece each. • Trainer Board-1 piece
- Connecting Wires

CIRCUIT DIAGRAM:

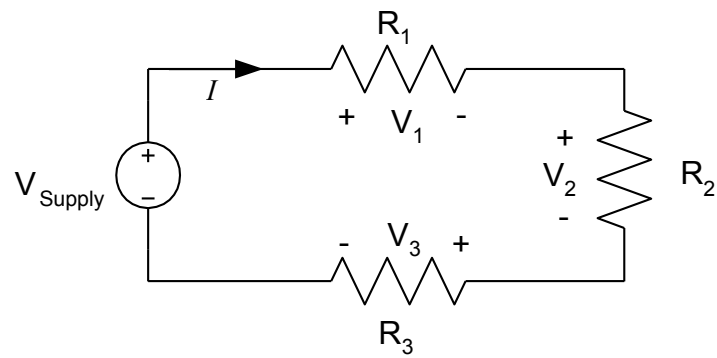


Figure 2.1

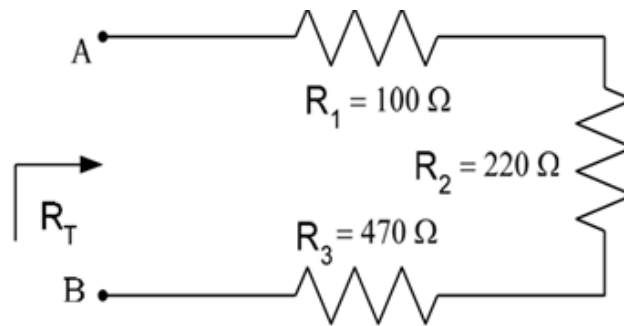


Figure 2.2

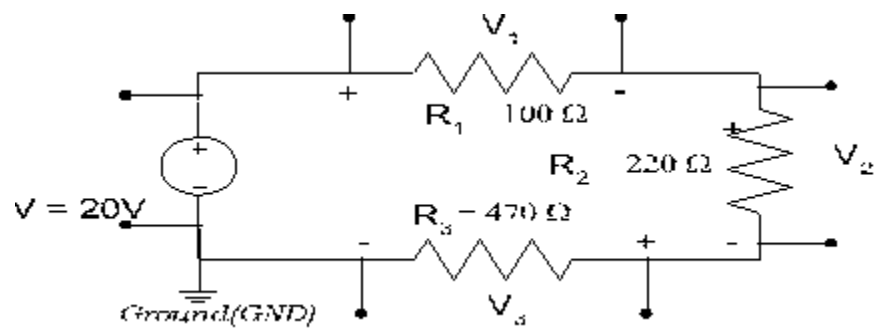


Figure 2.3

DATA SHEET:**Table 2.1**

Nominal values of Resistance (Ω)	Measured values of Resistance by Ohmmeter (Ω)	Equivalent Resistance, R_T		Measured voltage across each resistor (V)	Calculated Voltage using VDR (V)
		Measured R_T by using Ohmmeter (Ω)	Calculated $R_T = R_1 + R_2 + R_3$ (Ω)		
R1=100				V1=	
R2=220				V2=	
R3=470				V3=	

PROCEDURE:

1. Measure the resistances having values 100 Ω , 220 Ω & 470 Ω by using Ohmmeter and record the values in Table 2.1.
2. Construct the circuit as shown in Fig 2.2.
3. Then measure input resistance R_T across points A-B using Ohmmeter and record that value in Table 2.1.
4. Now construct the circuit as shown in Fig 2.3. Turn on the DC power supply and set the DC supply to 20V by using Voltmeter.
5. Measure voltage across each resistor with Voltmeter and record in the Table 2.1.
6. Calculate V1, V2 and V3 using Voltage Divider Rule (VDR). [Use measured values of resistances for all calculations.]

ASSIGNMENTS:

1. What can you deduce about the characteristics of a series circuit from observation Table 2.1?
2. From the data found in Table 2.1, mathematically prove that the current in the series network of figure 2.3 is equal for each resistance.
3. Verify KVL from the data obtained in Table 2.1.

THEORY:

In a parallel circuit (Figure 2.4) the voltage across parallel elements is the same.

The total or equivalent resistance (R_T) is given by,

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}$$

If there are only two resistors in parallel, it is more convenient to use,

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

In any case, the total resistance will always be less than the resistance of the smallest resistor of the parallel network.

KCL states that the currents entering a node must be equal to the currents leaving that node. For the network of Figure 2.4 the currents are related by the following expression:

$$I_T = I_1 + I_2 + I_3 + \dots + I_N$$

Applying current divider rule (CDR) for a circuit of only two resistors in parallel as shown in figure 2.5,

$$I_1 = \frac{R_2 I_T}{R_1 + R_2}, \quad I_2 = \frac{R_1 I_T}{R_1 + R_2}$$

For equal parallel resistors, the current divides equally and the total resistance is the value of one divided by the 'N' number of equal parallel resistors, i.e.:

$$R_T = \frac{R}{N}$$

For a parallel combination of N resistors, the current I_1 through R_1 is:

$$I_1 = I_T \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}}$$

EQUIPMENT:

- Variable DC power supply -1 piece
- Digital Multimeter (DMM)/ Analog multimeter-1 piece.
- Resistances: $1\text{K}\Omega$, $2.2\text{K}\Omega$, $4.7\text{K}\Omega$ -1 piece each.
- Trainer Board-1 piece
- Connecting Wires.

CIRCUIT DIAGRAM:

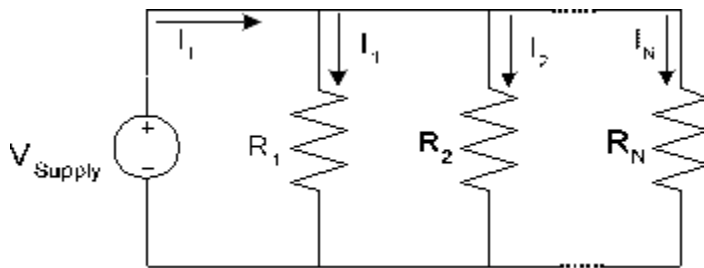


Figure: 2.4

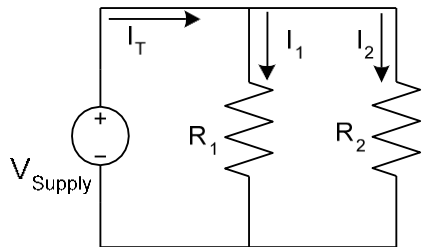


Figure: 2.5

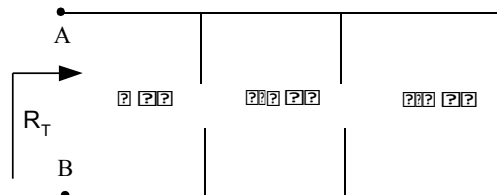


Figure: 2.6

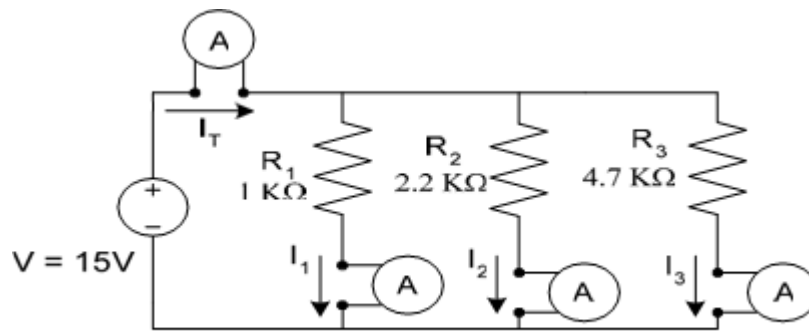


Figure: 2.7

PROCEDURE:

1. Measure the resistances having values $1\text{ K}\Omega$, $2.2\text{ K}\Omega$ & $4.7\text{ K}\Omega$ by using Ohmmeter and record the values in Table 2.2.
2. Construct the circuit as shown in Fig 2.6.
3. Then measure input resistance R_T across points A-B using Ohmmeter and record that value in Table 2.2.
4. Now construct the circuit as shown in Fig 2.7. Turn on the DC power supply and set the DC supply to 15V by using Voltmeter.
5. Measure the currents I_T , I_1 , I_2 and I_3 by using Ammeter and record in the Table 2.4.
6. Calculate I_1 , I_2 and I_3 using Current Divider Rule (CDR). [Use measured values of resistances for all calculations.]

DATA SHEET:

Table: 2.2

Nominal values of Resistance (K Ω)	Measured values of Resistance by Ohmmeter (K Ω)	Equivalent Resistance, R _T		Measured current through each resistor (A)	Calculated Current using CDR (A)
		Measured R _T by using Ohmmeter (K Ω)	Calculated $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ (K Ω)		
R ₁ =1				I ₁ =	
R ₂ =2.2				I ₂ =	
R ₃ =4.7				I ₃ =	

ASSIGNMENTS:

1. What can you deduce about the characteristics of a parallel circuit from observation Table 2.2?
2. From the data found in Table 2.2, Calculate I₁, I₂ and I₃ using Ohm's Law.
3. Verify KCL from the data obtained in Table 2.2.

Experiment No: 03**Name of the Experiment: Methods of Measuring Resistance Looking Through any Two Terminals of a Network.****OBJECTIVE:**

In this experiment we will learn the different methods used to measure the resistance looking through any two terminals of a circuit. In practical cases, it is not always possible to measure the resistance using an ohmmeter or by using circuit-reduction techniques. In many cases, especially in circuits containing electronic devices or dependent sources, indirect approaches are usually used to measure the resistance. We will be familiar with several such methods of measuring resistance. Besides, some more Spice syntax and commands will be introduced in this experiment.

EQUIPMENT:

- Trainer board-1 piece.
- Digital multimeter (DMM)-1 piece.
- Potentiometer: 10 K Ω -1 piece.
- Resistances:
 - 1 K Ω -4 pieces,
 - 2.2 K Ω -2 pieces,
 - 3.3 K Ω -3 pieces, 4.7 K Ω -2 pieces 10 K Ω -1 piece.

CIRCUIT DIAGRAM:

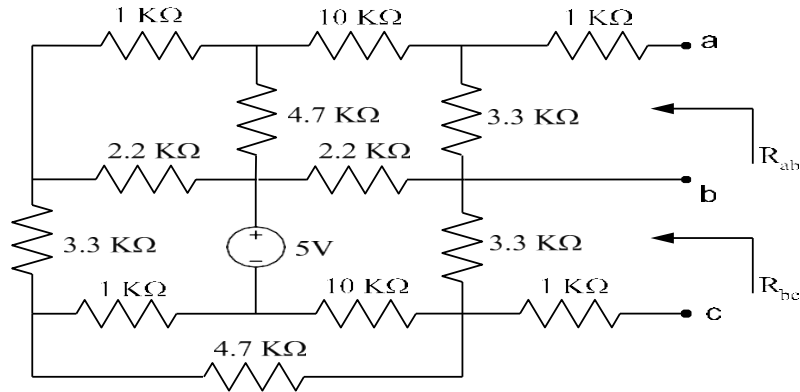


Figure 3.1: Resistance measurement

PROCEDURES:

Connect the circuit given in Figure 3.1. Then measure the equivalent resistance looking through terminals a-b (R_{ab}) and also the equivalent resistance looking through terminals b-c (R_{bc}). For this purpose, different methods will be used as described below:

Table 3.1: Measured resistances using different methods:

Resistance to be measured	Using Ohmmeter (K Ω)	Using V_s/I_s (K Ω)	Using VOC/ISC (K Ω)	Using Potentiometer (K Ω)
Rab				
Rbc				

(A) USING OHMMETER:

1. As the ohmmeter itself employs a battery to measure the resistance in a circuit, we have to de-energize the circuit or deactivate all the independent sources (i.e., we have to switch all the power supplies off) before measuring any resistance. This method is not applicable if the circuit contains any dependent source.

- Next we have to replace all the independent sources by their internal resistance. The resistance of an ideal voltage source is zero and the resistance of an ideal current source is infinite. So, the 5V voltage source in the circuit of Figure 3.1 should be removed completely and replaced by a short circuit (zero ohms). This will result in the circuit of Figure 3.2.

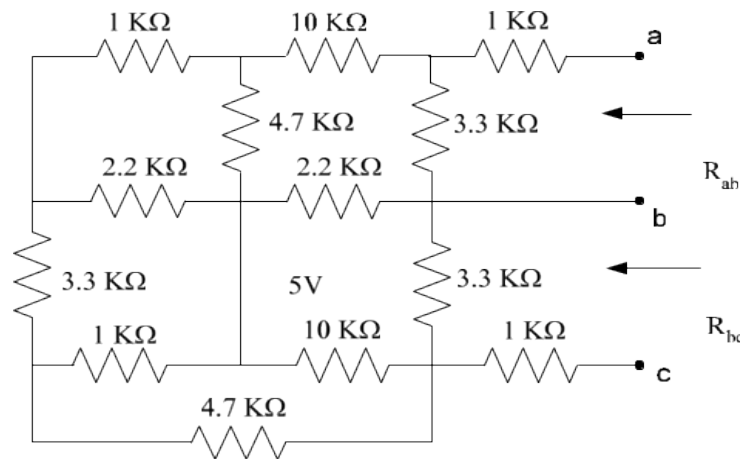


Figure 3.2: Resistance measurement using ohmmeter method

- Now to measure the resistance R_{ab} (equivalent resistance looking through terminals a-b), hold the ohmmeter terminals at terminals a and b and record the reading in Table 3.1.
- Similarly measure the resistance R_{bc} by holding the ohmmeter at terminals b and c and record the reading in Table 3.1.

(B) USING VS/IS:

- This method is an indirect approach of the above method. For this purpose, deactivate the 5V supply of Figure 3.1 as before and replace it with a short circuit.
- Now to measure the resistance R_{ab} , insert a voltage source V_s (take $V_s = 5V$) between terminals a and b as shown in Figure 3.3.
- Measure V_s and I_s (the current supplied by V_s) and calculate $R_{ab} = \frac{V_s}{I_s}$

4. To measure R_{bc} repeat step 2 (in this case, insert V_s between terminals b and c) and step 3. Record the readings in Table 3.1.

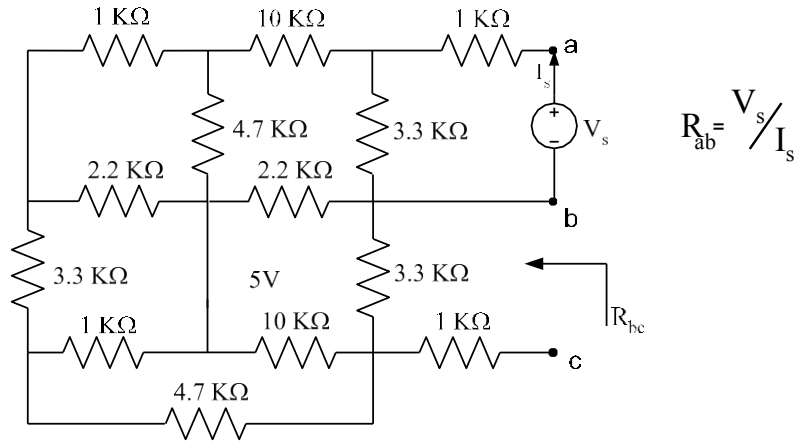


Figure 3.3: Resistance measurement using V_s/I_s method

(C) USING V_{OC}/I_{SC} :

1. This method uses Thevenin's theorem. Here the source need not be removed or deactivated as in previous two methods. This method is very useful in cases where the resistance is needed to be measured without deactivating the sources as in circuits containing dependent sources or electronic devices. Reconnect the circuit of Figure 3.1.
2. Measure the voltage V_{ab} in between terminals a and b. this voltage is called the open-circuit voltage V_{oc} .
3. Short-circuit the terminals a and b as shown in Figure 3.4 and measure the current flowing through these terminals. This is the short-circuit current, I_{sc} .
4. Calculate $R_{ab} = \frac{V_{oc}}{I_{sc}}$ and record this reading in Table 3.1.
5. Repeat steps 2 to 4 to measure R_{bc} in between terminals b and c.

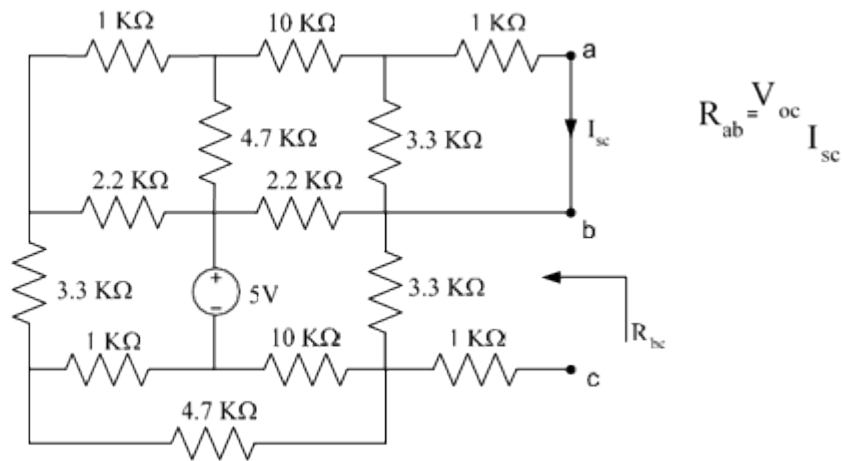


Figure 3.4: Resistance measurement using $\frac{V_{oc}}{I_{sc}}$ method.

(D) USING POTENTIOMETER:

1. This method also uses the principle of Thevenin's theorem and can be applied without removing any source from the original circuit. For this purpose, first measure the open-circuit voltage V_{oc} across terminals a and b of Figure 3.1.
2. Now connect a potentiometer in between terminals a and b as shown in V_{oc} Figure 3.5. Vary the pot until the voltage, V_{ab} .
3. Carefully remove the pot from the circuit and measure its resistance. The resistance of the potentiometer gives the equivalent resistance R_{ab} .
4. To measure the resistance R_{bc} , repeat steps 1 to 3 (in this case, terminals b and c should be used). Record these readings in Table 3.1.

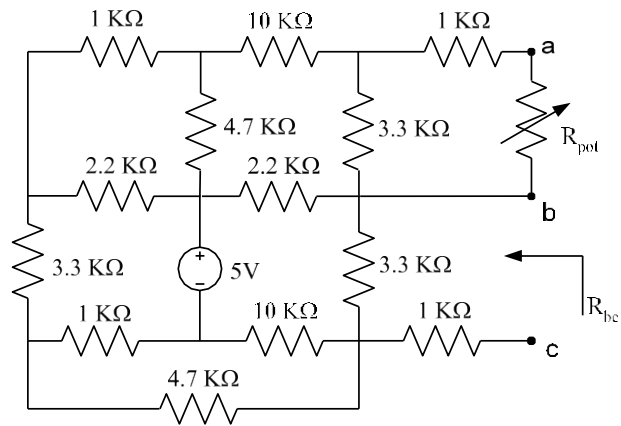


Figure 7.5: Resistance measurement using potentiometer method